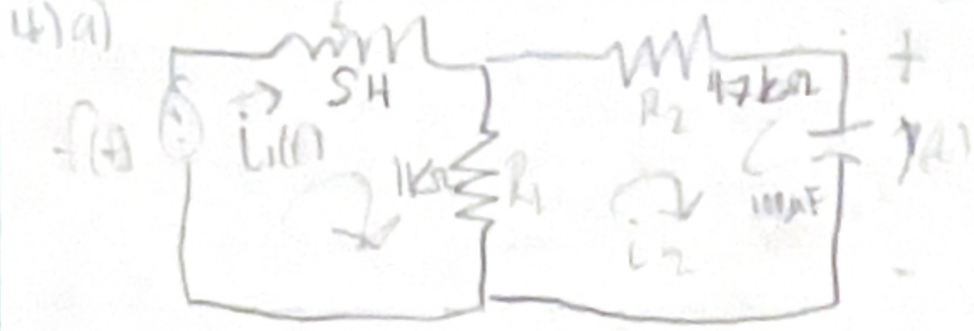
4) a)



loop 1: $f(t) = L\frac{di_1}{dt} + R_1(i_1 - i_2)$

loop 2: $0 = R_1(i_2 - i_1) + R_2 i_2 + y(t)$

$= R_1 i_2 - R_1 i_1 + R_2 i_2 + y(t)$

$R_1 i_1 = R_1 i_2 + R_2 i_2 + y(t)$

$i_1 = i_2 + \frac{R_2 i_2 + y(t)}{R_1}$

$i_2 = CDy(t)$

Plug in i_1

$f(t) = \frac{LDy(t)}{R_1} + \frac{LDR_2 i_2}{R_1} + LDi_2 + y(t) + R_2 i_2 + R_2 R_1 i_2$

$f(t) = \frac{LCR_2 D^2 y(t)}{R_1} + LCD^2 y(t) + \frac{LDy(t)}{R_1} + y(t) + R_2 CDy(t)$

$f(t) = \left(\left(\frac{LCR_2}{R_1} + LC\right)D^2 + \left(\frac{L}{R_1} + R_2 C\right)D + 1\right) y(t)$

$f(t) = (.024D^2 + 4.705D + 1)\, y(t)$

b) $0 = y(t) + R_2 i_2 + R_1 i_2 - R_1 i_1$

$R_1 i_1 - y(t) = (R_2 + R_1) i_2$

$i_2 = \frac{R_1 i_1 - y(t)}{R_2 + R_1}$

$\frac{dy(t)}{dt} = \frac{i_2}{C}$

$\frac{dv(t)}{dt} = \frac{R_1 i_1 - y(t)}{R_2 + R_1}$

$\frac{dv(0)}{dt} = 40.625$

c)

$(.024D^2 + 4.705D + 1)\, y(t) = 0$

$(D^2 + 196.04D + 41.667)\, y(t) = 0$

$\lambda = \frac{-196.04 \pm \sqrt{196.04^2 - 4(41.667)}}{2}$

$(\lambda + 0.2128)(\lambda + 195.83)\, y(t) =$

$\lambda = -0.2128 \quad \& \quad -195.83$

$y(t) = C_1 e^{-0.2128t} + C_2 e^{-195.83t}$

$y'(t) = -0.2128 C_1 e^{-0.2128t} + -195.53 C_2 e^{-195.83t}$

$y(0) = 5 = C_1 + C_2$

$y'(0) = 40.625 = -0.2125 C_1 - 195.53 C_2$

$\begin{bmatrix} 1 & 1 & 5 \\ -0.2128 & -195.53 & 40.625 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & 5.2134 \\ 0 & 1 & -0.2134 \end{bmatrix}$

$y(t) = 5.2134 e^{-0.2128t} - 0.2134 e^{-195.53t}$